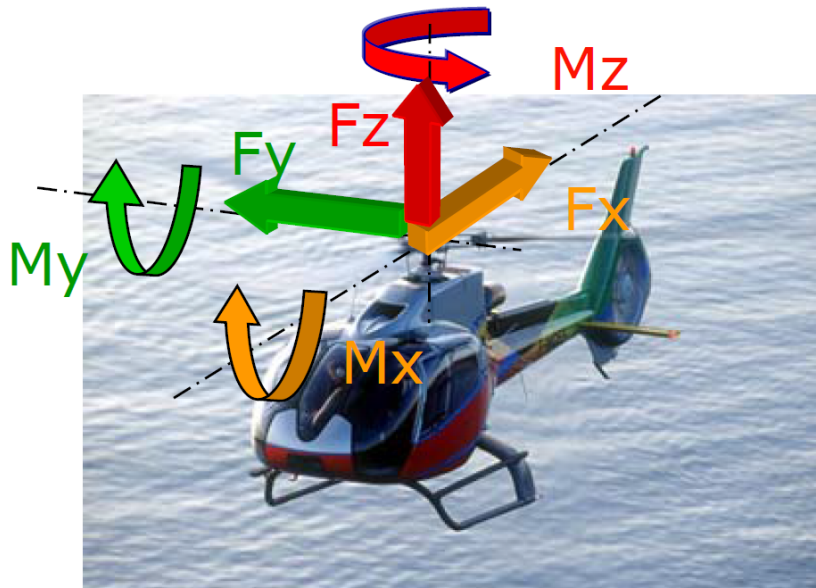


Helicopter Thrust force calculation BEM, forward flight

Equations of motion in a 3-dof model

Dynamic equations of motion in a longitudinal plane



$$\dot{u} = -g \sin \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - qw$$

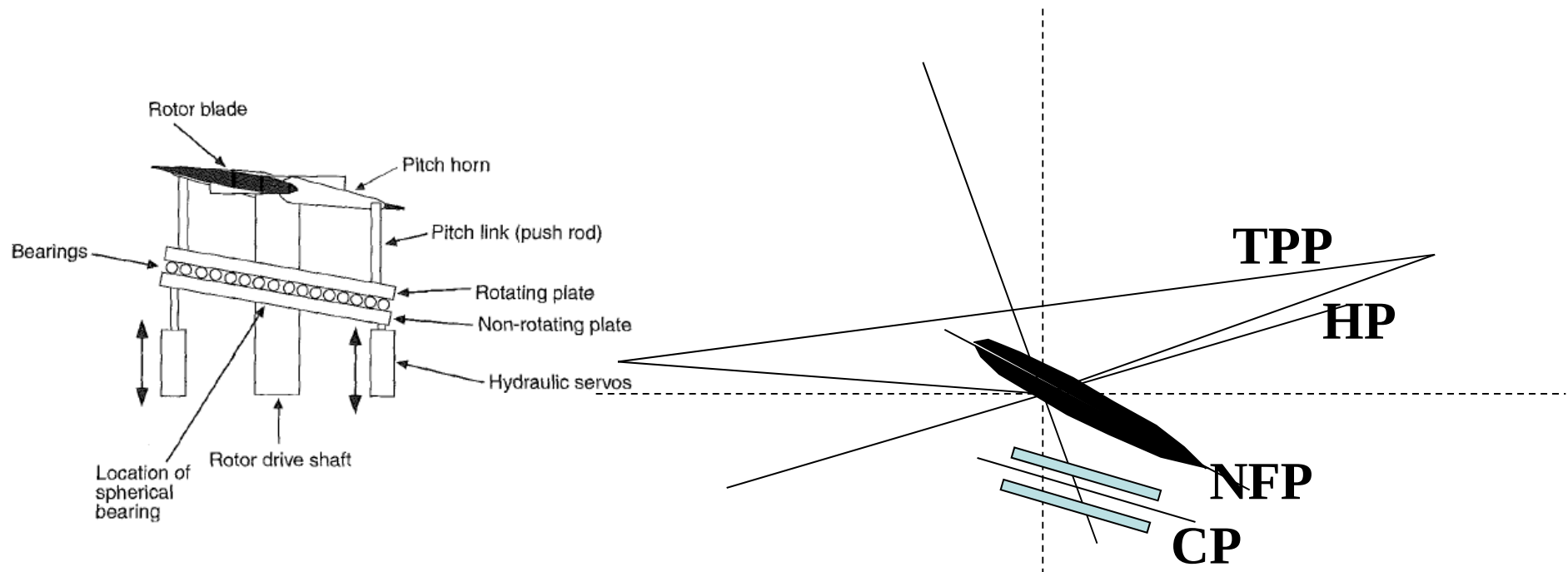
$$\dot{w} = g \cos \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + qu$$

$$\dot{q} = -\frac{T}{I_y} h \sin(\theta_c - a_1)$$

$$\dot{\theta}_f = q$$

How to calculate the thrust force?

Main rotor reference systems



TPP (DP) = tip-path-plane (disc plane) is *the plane formed by the blade tips*

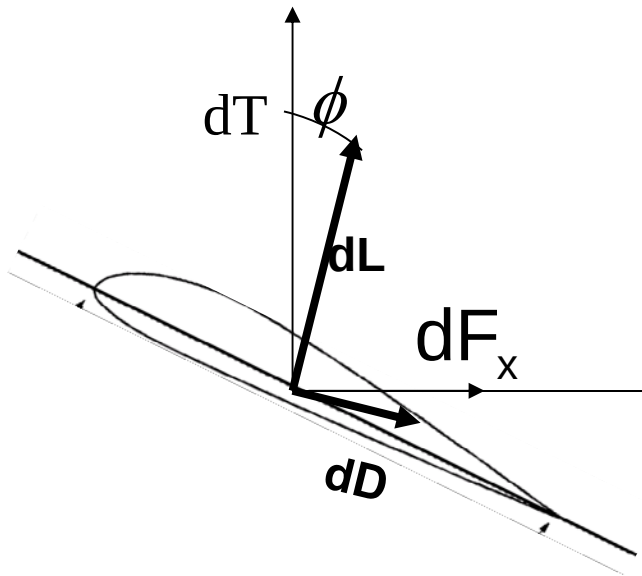
SP (HP) = shaft plane (hub plane) is *the plane perpendicular to the hub*

NFP = no-feathering plane is *the plane where an observer sees no variation in cyclic pitch*

CP = control plane (swashplate plane) is *the plane containing the swashplate*

Thrust force in BEM

THIS DEMONSTRATION CONSIDERS THE CONTROL PLANE AS
PLANE OF REFERENCE FOR THE ROTOR

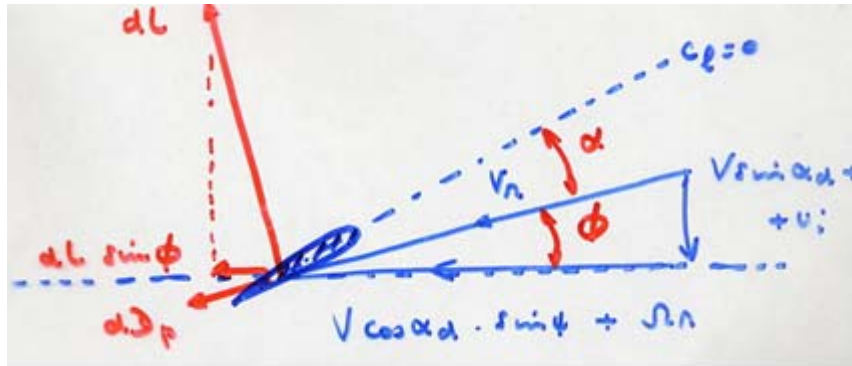


$$dT = dL \cos \phi - dD_p \sin \phi \simeq dL \quad (6)$$

But:
$$dL = \frac{1}{2} \rho (c dr) V^2 C_{l\alpha} \alpha \quad (7)$$

Thrust force in BEM

What is the angle of attack and velocity on the blade element?



$$V \approx \Omega r + V \cos \alpha_c \sin \psi \quad (8)$$

$$\alpha = \theta - \phi = \theta - \frac{V \sin \alpha_c + v_i}{\Omega r + V \cos \alpha_c \sin \psi} \quad (9)$$

Substituting (8) and (9) into (7) gives:

$$dL \approx \frac{1}{2} \rho C_{l\alpha} [\Omega r + V \cos \alpha_c \sin \psi]^2 \left(\theta - \frac{V \sin \alpha_c + v_i}{\Omega r + V \cos \alpha_c \sin \psi} \right) c dr \quad (10)$$

Thrust force in BEM

Now, integrating the lift force along the radius and azimuth gives the total thrust force of the rotor as:

$$T = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R C_{l\alpha} \frac{1}{2} \rho [\Omega r + V \cos \alpha_c \sin \psi]^2 \left(\theta - \frac{V \sin \alpha_c + v_i}{\Omega r + V \cos \alpha_c \sin \psi} \right) c dr \quad (11)$$

The following non-dimensional notations are made:

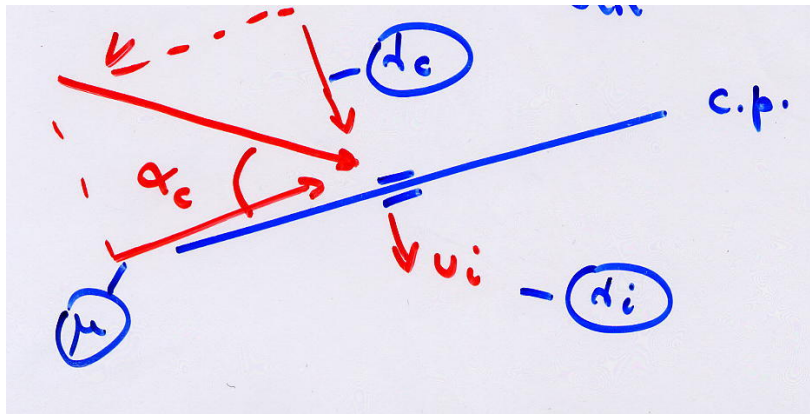
$$C_T = \frac{T}{\rho (\Omega R)^2 \pi R^2}$$

Thrust coefficient

(12)

Thrust force in BEM

Non-dimensional component of the velocities perpendicular to control plane:



$$\lambda_c = \frac{V \sin \alpha_c}{\Omega R} \quad (13)$$

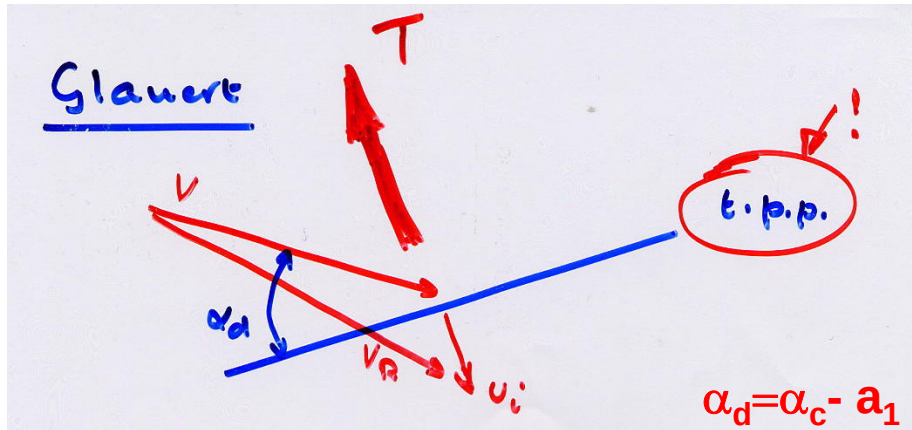
$$\lambda_i = \frac{V_i}{\Omega R} \quad (14)$$

Substituting (12) to (14) into the integral (11) and performing the integration gives finally the thrust coefficient in BEM as:

$$C_{T \text{ elem}} = \frac{C_{l\alpha} \sigma}{4} \left[\frac{2}{3} \theta_0 \left(1 + \frac{3}{2} \mu^2 \right) - (\lambda_i + \lambda_c) \right] \quad (16)$$

Thrust force in Glauert theory

This Demonstration Considers The Tip Path Plane As Plane Of Reference For The Rotor



$$T = \dot{m} \cdot 2v_i \quad (17)$$

But:

$$\dot{m} = \rho \pi R^2 V_R =$$

$$\rho \pi R^2 \sqrt{(V \sin \alpha_d + v_i)^2 + V \cos \alpha_d^2} \quad (18)$$

Substituting (18) into (17) gives:

$$T = 2\rho\pi R^2 v_i \sqrt{(V \sin \alpha_d + v_i)^2 + V \cos \alpha_d^2} \quad (19)$$

Thrust force in Glauert theory

Using the definition of thrust coefficient (12) and the non-dimensional notations (13) and (14) into expression (19) and observing that the angle of attack in the disc plane is $\alpha_d = \alpha_c - a_1$ gives the expression of thrust coefficient in Glauert theory as:

$$C_{T\,Gl} = 2\lambda_i \sqrt{\left[\frac{V}{\Omega R} \sin(\alpha_c - a_1) + \lambda_i \right]^2 + \left[\frac{V}{\Omega R} \cos(\alpha_c - a_1) \right]^2} \quad (20)$$